



May 2025

ME2004 -1 - TECHNOLOGY AND PROCESSES IN HEAVY MANUFACTURING
(For ME)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- Which of the following is not included in metal forming process?
A) Rolling
B) Forging
C) Extrusion
D) Power metallurgy
- Tungsten carbide (WC) is a product as a cutting tool are manufactured from _____.
A) Powder metallurgy
B) Casting
C) Forging
D) Rolling
- In which type of Ashby's chart shows, the stress measures the resistance to failure of a material
A) Young's Modulus – Density
B) Strength – Density
C) Strength – Toughness
D) Strength – Elongation
- Which property is a useful for designing heat sinks and insulating components.
A) Coefficient of Thermal Expansion
B) Thermal Conductivity
C) Compressive Strength
D) Melting point
- Two cubical castings made from the same metal, one with a side length of 5 cm and the other with a side length of 10 cm, are made using green sand. If the smaller casting takes 4 minutes to solidify, how long will the larger casting take to solidify?
A) 4 min
B) 8 min
C) 16 min
D) 20 min
- A converging passage is used to direct the flow of liquid molten metal into the mold.
A) Increase the rate of feeding
B) Quickly break off the protruding portion of the casting
C) Decrease wastage of cast metal
D) Avoid aspiration of air
- A very thin and complex geometries is obtained using _____.
A) Roll forging
B) Isothermal forging
C) Radial forging
D) Hot-Die Forging
- _____ is a main factor in precision forging
A) Material
B) Cost
C) Machining
D) Design
- The following is not related to drawing force in deep drawing process.
A) Drawing ratio
B) Yield Tensile strength
C) Strength of punch material
D) Diameter of the punch

10. Fiber elongation (FE) in warm rolling is in the range of _____
 A) $3 \leq FE \leq 6$ B) $5 \leq FE \leq 9$
 C) $FE < 5$ D) $FE > 9$
11. Which of the following defect results from electrode coating and the flux used?
 A) Porosity B) Cracks and micro-fissures
 C) Slag inclusion D) Hot cracking
12. Which of the following is not with respect to Gas Metal Arc Welding (GMAW)?
 A) A non-consumable electrode B) The process is carried out without application of pressure
 C) Protective gas can be used D) The torch assembly includes the gas supply device and the charging wire
13. Which of the following welding method have least heat affected zone?
 A) Plasma arc welding B) Laser beam welding
 C) Electron beam welding D) Gas metal arc welding
14. Which of the following statement is not correct in Thermite welding process?
 A) Thermite welding is Exothermic process B) Heat is generated as a result of a chemical reaction between aluminum and oxide
 C) The burning of iron and aluminum produces heat of up to 3000° F D) Thermite welding is also used to repair forgings and steel castings
15. The joint formed by Metal diffusion and metal-to-metal bond formation are accelerated by the high-pressure force is known as _____.
 A) Resistance welding B) Laser beam welding
 C) Friction stir welding D) Electron beam welding
16. Which of following cutting tool material provides higher cutting speed?
 A) Coated carbide B) Cermets
 C) High speed steel D) Ceramics
17. Which of the wear is the indicator for Tool life?
 A) Crater wear B) Flank wear
 C) Corner wear (or) Nose wear D) Built-up- edge
18. When the power is transmitted through Oil based system, then the system is termed as
 A) Fluid power system B) Pneumatic system
 C) Hydraulic system D) Electro hydraulic system
19. Which machine cuts by rotating a blade back and forth across a fixed die in a series of up and down punching motions?
 A) Nibblers B) Snips
 C) Throatless shear D) Hydraulic punch
20. What is the most important water property in waterjet machining?
 A) Density B) Viscosity
 C) Compressibility D) Velocity

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the Primary and secondary manufacturing processes.	4	L*2	1	1
	b) Describe the closed die forging operation.	4	L3	2	2
	c) Explain the basics steps involved in refining.	4	L2	1	1
	d) What is the purpose of designing a tapered sprue in casting, and how does it prevent the aspiration effect?	4	L4	2	2

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| 2. a) | What is the significance of mechanical properties such as tensile strength, hardness, and ductility in material selection for engineering applications? | 4 | L4 | 1 | 2 |
| b) | Explain the basic bulk deformation process. | 4 | L2 | 2 | 1 |
| c) | Explain the various equipment used in the power plant industry. | 4 | L2 | 1 | 1 |
| d) | What principle governs the solidification time of a casting, and how is it related to the casting size? | 4 | L3 | 2 | 2 |
| 3. a) | Describe the formation of carbides and nitrides in alloy. | 4 | L2 | 1 | 1 |
| b) | A metal cube with a side length of 3 cm is cast in green sand and solidifies in 3 minutes. If a similar cube with a side length of 6 cm is cast, how long is it expected to take to solidify? | 4 | L3 | 2 | 2 |
| c) | What are the key operations involved in primary and secondary manufacturing processes, and why are they significant? | 4 | L4 | 1 | 2 |
| d) | A down sprue with a length of 240 mm and a top diameter of 30 mm has a molten metal height of 80 mm in the pouring cup. Calculate the required diameter of the sprue at its lower end to prevent the occurrence of aspiration effect during pouring. | 4 | L3 | 2 | 3 |

Unit – II

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|-------|---|---|----|---|---|
| 4. a) | Describe the angular limits for various types of joints:
i) Edge Joint ii) Corner Joint iii) Butt Joint iv) Lap Joint. | 4 | L2 | 3 | 1 |
| b) | Describe the theory and types of chip formation in metal machining. | 4 | L4 | 4 | 2 |
| c) | Explain the shielded metal arc welding process and its advantages. | 4 | L2 | 3 | 1 |
| d) | What is the difference between drilling and boring operations in machining? | 4 | L3 | 4 | 2 |
| 5. a) | Describe the Submerged Arc Welding (SAW) process and explain how flux influences the weld. | 4 | L2 | 3 | 1 |
| b) | What is Taylor's tool life equation, and how is it used to predict tool life in machining? | 4 | L3 | 4 | 2 |
| c) | What is Electron Beam Welding (EBW), and how does it work? | 4 | L2 | 3 | 1 |
| d) | Applying Taylor's tool life equation with an exponent $n=0.5$, determine the ratio of the new tool life to the original when the cutting speed is reduced by half. | 4 | L4 | 4 | 2 |
| 6. a) | Describe the Gas Tungsten Arc Welding (GTAW) process and explain how the tungsten electrode and inert gas affect weld quality. What are its key benefits? | 4 | L2 | 3 | 1 |
| b) | What are the key differences between turning, shaping, and planning operations? | 4 | L4 | 4 | 2 |
| c) | What factors influence the selection of a particular weld joint in different applications? | 4 | L3 | 3 | 2 |
| d) | What is the grinding operation in machining, and what are its main applications? | 4 | L2 | 4 | 1 |

Unit – III

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|-------|--|---|----|---|----|
| 7. a) | Describe the key components and purpose of a quality control plan. | 4 | L2 | 5 | 6 |
| b) | Define the Design for Inspection (DFI) and explain its significance in product design and manufacturing. | 4 | L4 | 5 | 2 |
| c) | Describe the basic components and working principles of hydraulic systems. | 4 | L2 | 5 | 1 |
| d) | Define industry 4.0. | 4 | L1 | 5 | 12 |

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|-------|---|---|----|---|----|
| 8. a) | Explain the classification of pressure vessel based on geometry, orientation and purpose. | 4 | L2 | 5 | 1 |
| b) | Explain the component of pressure vessel with respect to shell, head, nozzle and support. | 4 | L2 | 5 | 1 |
| c) | Describe the basic components and working principles of pneumatic systems. | 4 | L2 | 5 | 1 |
| d) | Explain the primary drivers of industry 4.0. | 4 | L4 | 5 | 12 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
